PAPER_707: NGC 2525 Barred Spiral (Variant 2): SN Ia Feedback UQFF Analysis

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Abstract

NGC 2525 is an SBc barred spiral galaxy at $z = 0.0051$ (721 Mly), host of Type Ia supernova SN 2018gv discovered during Hubble's Frontier Fields program. This variant-2 analysis derives the UQFF MUGE with explicit SN Ia feedback impulse and bar streaming electromagnetic contributions.

1. System Parameters

Parameter	Value	Description
$M_{stellar}$	$3 \times 10^{10} M_\odot$	Stellar mass
M_{DM}	$9 \times 10^{10} M_\odot$	Dark matter halo
r_{gal}	30 kpc	Galaxy radius
z	0.0051	Redshift
L_{SN}	10^{43} W	SN Ia peak luminosity
τ_{SN}	3.156×10^8 s	SN decay (~ 10 yr)

2. Master UQFF Gravity Equation

$$g_{v2}(t) = \frac{G(M_\star + M_{DM})}{r^2}(1+H_z t)(1+f_{TRZ}) + \frac{L_{SN}}{cM_{tot}r}\, e^{-t/\tau_{SN}} + q(v_{bar} \times B_{bar}) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right) \times 10^{-12}$$

2.1 SN Ia Feedback Impulse

$$a_{SN}(t) = \frac{L_{SN}}{cM_{tot}r_{gal}}\, e^{-t/\tau_{SN}}$$

The SN Ia releases $\sim 10^{44}$ J over 10 yr, providing a brief but measurable gravitational impulse correction to the MUGE.

3. References

- Yang et al. (2020), ApJ, 890, 177 (SN 2018gv in NGC 2525)
- UQFF Framework, Star-Magic Session 175

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